Training a CNN to estimate population density in satellite images

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Link to repository: https://github.com/manforowicz/satellite-image-population-counter

Question
Can a CNN (convolutional neural network) estimate population density in satellite images?

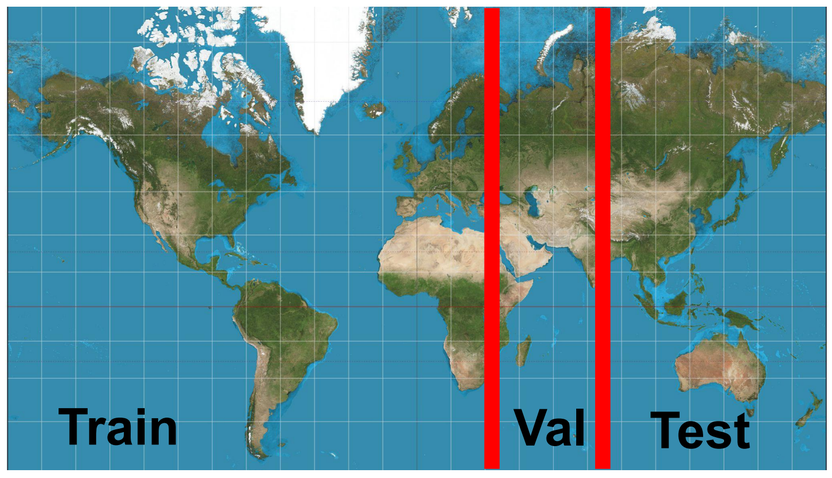
Similar deep learning methods can be applied to:
- Tracking disasters like floods and wildfires
- Calculating urban statistics like greenery coverage, paved area, housing construction, and km of streets.

Data from satellite imagery is more consistent than reports by jurisdictions with different methodologies.

Data preparation
WGS84 Population map from GHSL (Global Human Settlement Layer)[1], an open-source project by the European Commission. Each pixel value covers 3 arcsec (roughly 100m) and specifies how many people live in that region.
Satellite images were fetched from Google Static Maps API:
- 1km x 1km and 244 x 244 pixels
- Randomly weighted toward populated areas, to avoid a sparse dataset.
- Labelled with ln(1+average_density) as this makes the distribution less skewed, and more learnable. Calculated from averaged GHSL pixels.

Data Split
The dataset consists of 14,588 images, split based on longitude. This course-grained grouping lets us investigate whether it's possible to generalize from mostly the West Hemisphere, to the rest of the world.
- -180° ≤ longitude < 30°: Train (6823 images)
- 30° ≤ longitude < 80°: Validation (3587 images)
- 80° ≤ longitude ≤ 180°: Test (4178 images)

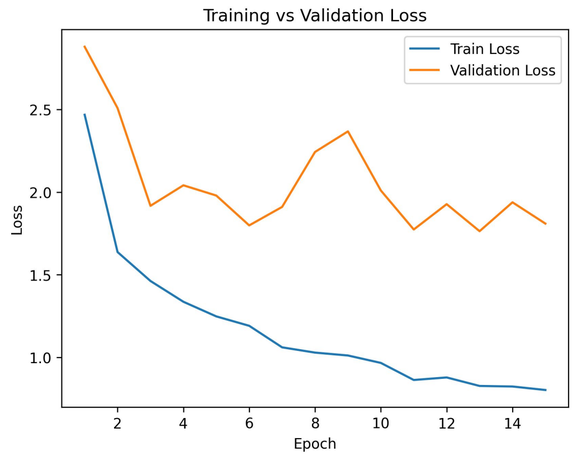
Training images are augmented through random flipping and varying brightness, contrast, saturation, and hue. I observed this very significantly decreased overfitting by artificially increasing dataset size,



Model architecture
The model is a ResNet-50[2] pre-trained on ImageNet with its final FC (fully-connected) layer replaced to output a single number – the log1p of the image's population density.

After open-ended experimentation, I used these hyperparameters:
- Learning rate: 4e-4 Number of epochs: 15
- Weight decay: 0 Dropout probability: 0
- Only backbone weights before layer 1 frozen.

Performance
MSE of log1p estimates:
- training: 0.8
- validation: 1.8
- test: 1.9



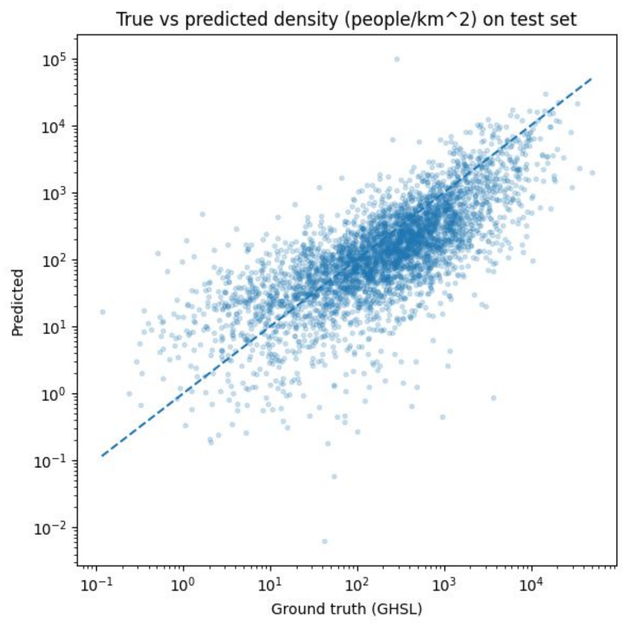
References
[1] GHSL (https://human-settlement.emergency.copernicus.eu/)
[2] Deep Residual Learning for Image Recognition (https://arxiv.org/pdf/1512.03385)

Related works
- A Deep Learning Approach for Population Estimation from Satellite Imagery
- Mapping Missing Population in Rural India: A Deep Learning Approach with Satellite Imagery
- Sensing Population Distribution from Satellite Imagery Via Deep Learning: Model Selection, Neighboring Effects, and Systematic Biases

Test images with best accuracy (<0.001 log1p MSE)
These images are typical urban, rural, or suburban environments. The model generalizes surprisingly well to different urban structures.

Test images on which the model overestimated population (>5 log1p MSE)
These images have natural features or mining operations that differ from the training region, and look like settlements. Also, some satellite images were incomplete, with black squares in them, which misled the model.

Test images on which the model underestimated population (>5 log1p MSE)
These images look like natural landscapes, but actually have accidentally well-camouflaged settlements. However, some of these images look genuinely unpopulated, which puts into question the accuracy of GHSL data, or the alignment of the satellite images to it.



Conclusion
- Model easily distinguishes between urban and rural areas, but fails to differentiate more closely. Predictions were off by an MSE log1p of 1.9. This translates to "*typically*" off by a factor of around 3.6.
- This is not great, but probably better than I could do. Population densities vary by orders of magnitude between human settlements.
- I learned that generalizing from the West Hemisphere to the rest of the world is possible, but yields mixed results. In a real environment, I'd train on images from all over the world.

Things to try in the future
- Predict population density on satellite images of different zoom, inputting scale information to the model.
- More than 14,588 images. The world is huge, and currently the model had significant overfitting (though data augmentation partially mitigated that).
- Different regional train, validation, and test split.
- More combinations of hyperparameters.